SL8521E Thermal Design Guide

HW

2018.07.11

TECHNOLOGY FOR THE MOBILE WORLD

紫光展锐 UNIGROUP SPREADTRUM & RDA

History

Version	Owner	Date	Notes
V1.0	Feihu.Ye	2018.07.11	First version

Contents

Overview

- Main heat source components of SL8521E platform:

  AP – SL8521E

  Memory

  PMU – SC2721G

  2G/4G PA

  Transceiver – SR3593A

  LCD backlight

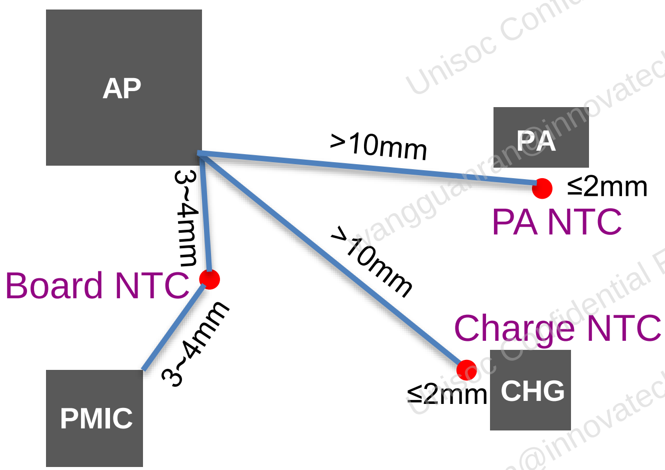
  Charge IC

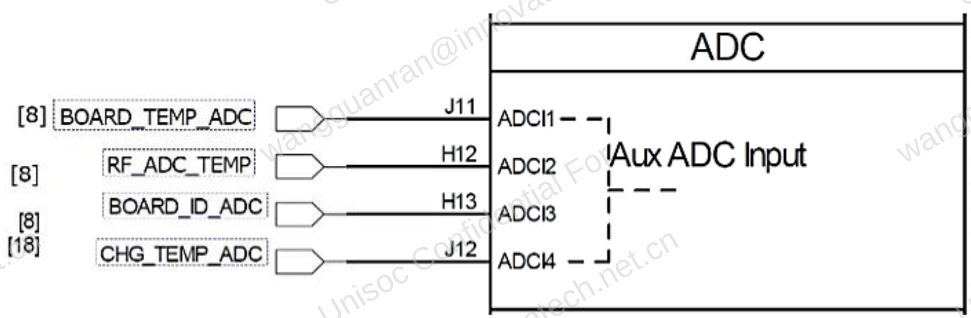
Thermal control architecture

- Thermal sensors
- Thermal management architecture

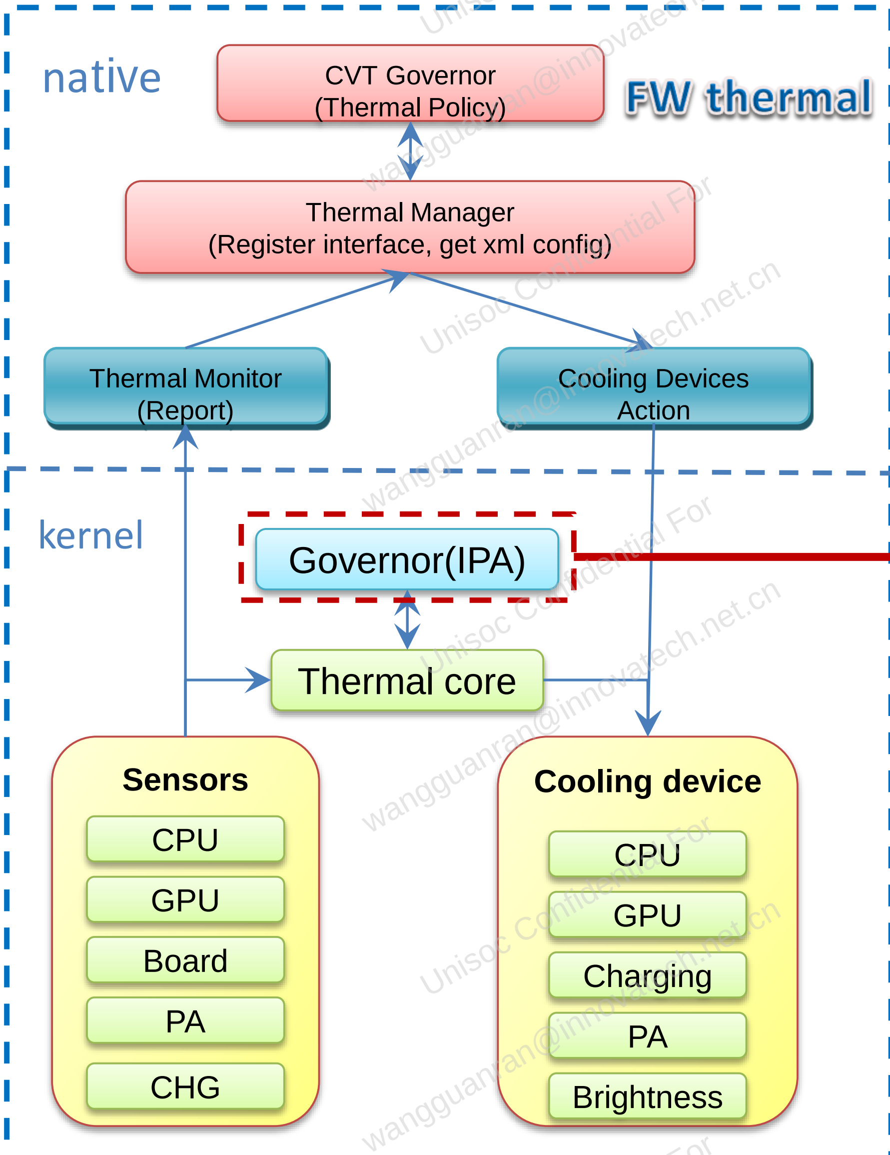
Thermal sensors

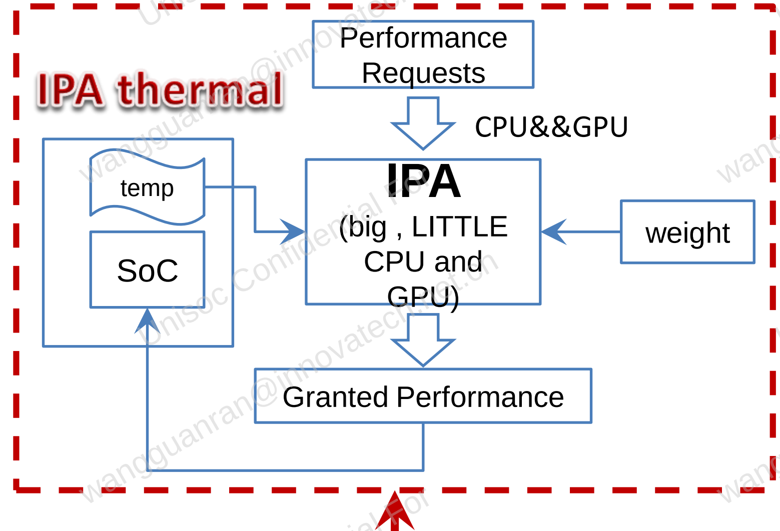
- Besides the on-die sensors in AP, additional on-board NTC (thermistors) are needed for system thermal management.
- NTC placement rules and ADC channel connect rules are shown as below.
- PA NTC monitors PA temperature. PA power back-off is expected to trigger by PA NTC. It’s critical for smartwatch to conduct modem thermal mitigation.





Thermal control architecture





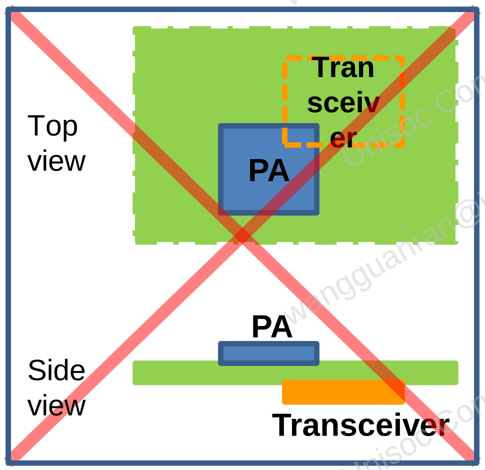
- IPA - Intelligent power allocation
  - IPA thermal control CPU & GPU
- FW - Framework
  - FW thermal control cooling devices in different scenarios

PCB thermal design notes

- Placement consideration
- Routing consideration
- TCXO consideration

PCB thermal design notes

- Placement consideration

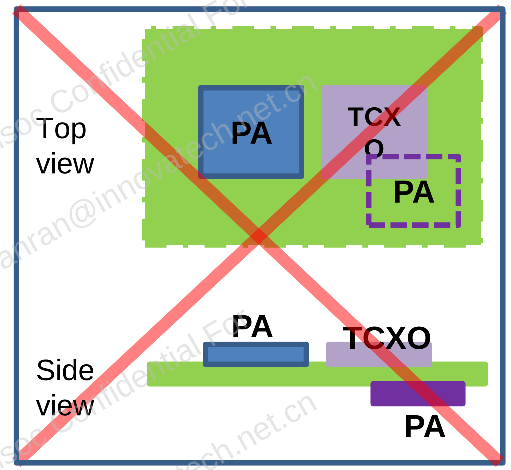


Not recommend

Case1: Heat sources placed side by side

Not recommend

Case2: Heat sensitive component (TCXO) placed too close to the heat source (PA / transceiver)

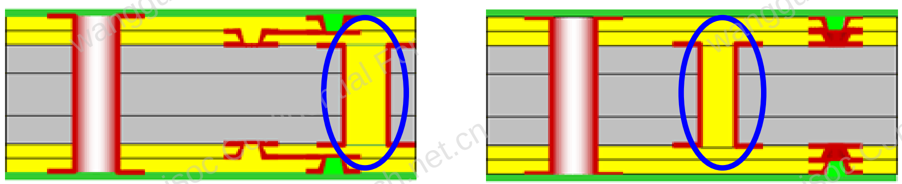


- PCB smaller than 2500mm^2 would have extreme high thermal risk. 3000mm^2 or larger is recommended to avoid high thermal risk.
- Heat source should keep away from board edge by 3mm at least.
- Heat source should keep away from each other by 5mm at least.

PCB thermal design notes

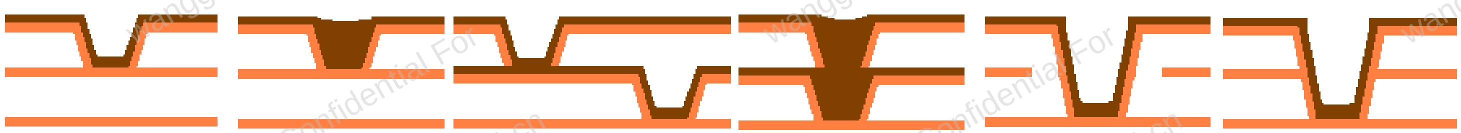
- Routing consideration
  - It is recommended to design buried via holes instead of blind via holes, with 10mil via diameter and 1mil plating thickness.

Buried Via:



Through via　stagger via　Through via　Stack via

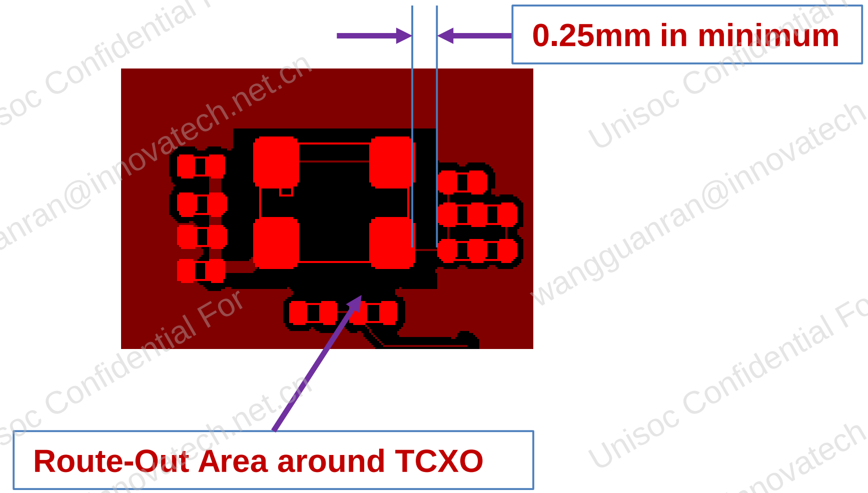
Blind Via:



Micro via hole　Solid via hole　stagger via hole　Stack via hole　Skip via hole　Step via hole

PCB thermal design notes

- TCXO/TSX consideration



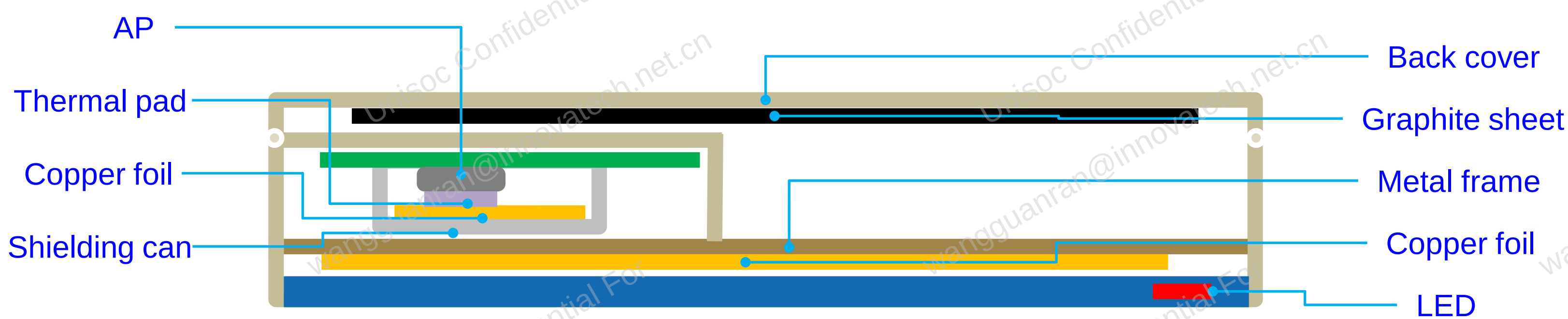
- ➢ Distance from TCXO/TSX to heat source should be 3mm/4~6mm separately at least.
- ➢ Minimum 0.25mm-clearance route out area around TCXO/TSX to avoid heat conductivity effects.
- ➢ Recommend no-copper area under TCXO/TSX from Layer L1 to Layer L(n-1).
- ➢ Keep ≥1mm away from the PCB edge, don't refer to the PCB Main GND, refer PCB design guide for detail.

Mechanical thermal design notes

- Total thermal solution
- Camera consideration
- Fingerprint module consideration
- Shielding can consideration
- Mechanical consideration

Mechanical thermal design notes

- Total thermal solution



➢Suggest to add metal frame between LCM and PCB.

➢TIM must be applied between AP and shielding can , K should be bigger than 2w/mk.

➢Consider to attach graphite sheet on back cover to cool down heat spot , K in x-y should better than 1300 w/mk.

➢Strongly suggest to place LCM LED and PCB on opposite position.

Mechanical thermal design notes

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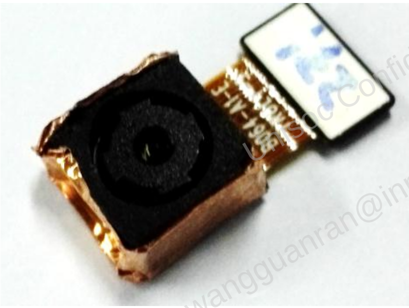
- Camera consideration



➢Suggest to surround camera with copper foil, and attach it to metal frame with thermal pad.

➢Suggest to use high efficiency LED as flash light and LCD backlight ,30% is preferred.

Mechanical thermal design notes

- Fingerprint module consideration

➢Suggest to mount fingerprint module to back cover, and connect to PCB by S sharp FPC, it could prevent heat spread to back cover.

➢Do not mount fingerprint module to PCB directly.

Thermal test notes

- Recommended test equipment
- Recommended test procedure
- Thermal debug procedure
- Thermal debug tool

Thermal test notes

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- Recommended test equipment

Test Equipment	Model	Figure	Function
Thermal chamber	BINDER KB-115-E3.1		Keep specified temperature
Infrared camera	FLIR T420		Identify hot spot
Data logger	YOKOGAWA GP10		Get accurate temperature curve
Thermal couple	Omega TT-K-36-SLE		Attach to hot spot and connect to date logger

Thermal test notes

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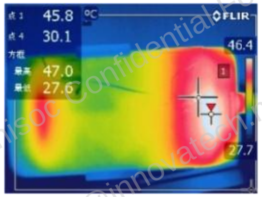
- Recommended test procedure

1.Run specified scenario by 30 min

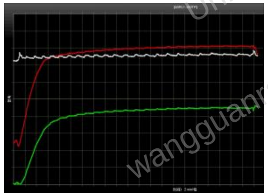
2.Take IR photo to get surface temperature



3.Identify hot spot to attach thermal couple

4.Connect thermal couple to data logger

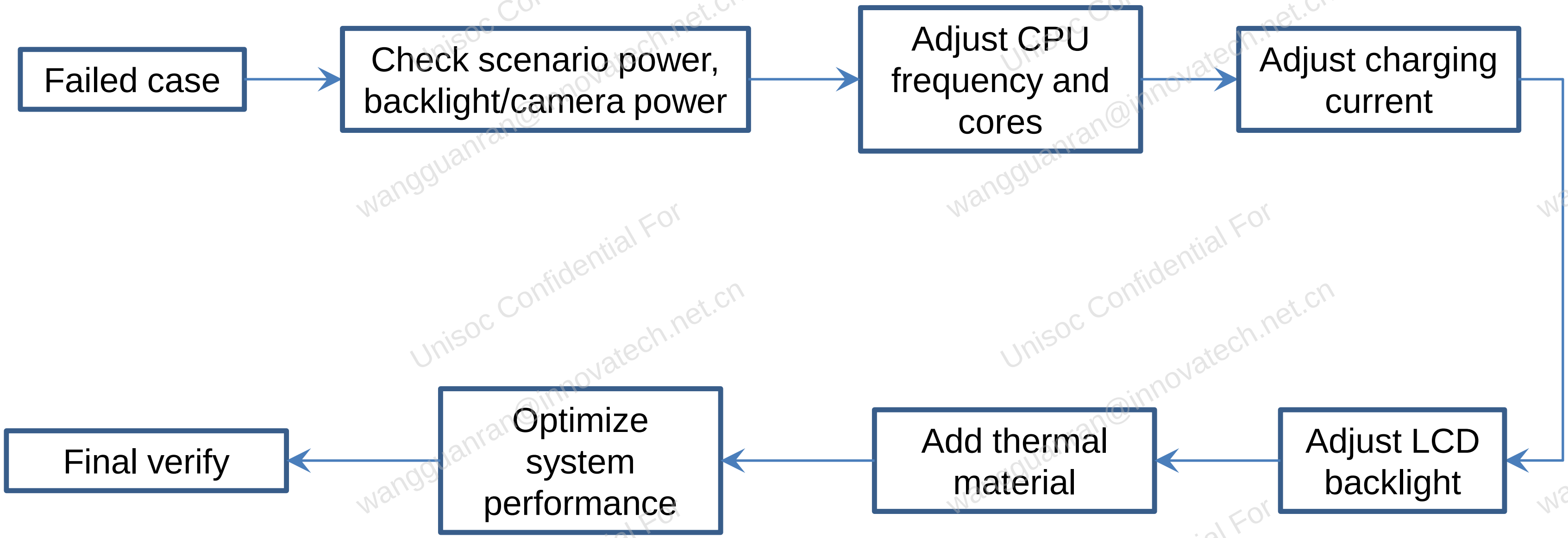
5.Put test phone into chamber and Keep specified temperature and still airflow



6.Log temperature curve changed with time

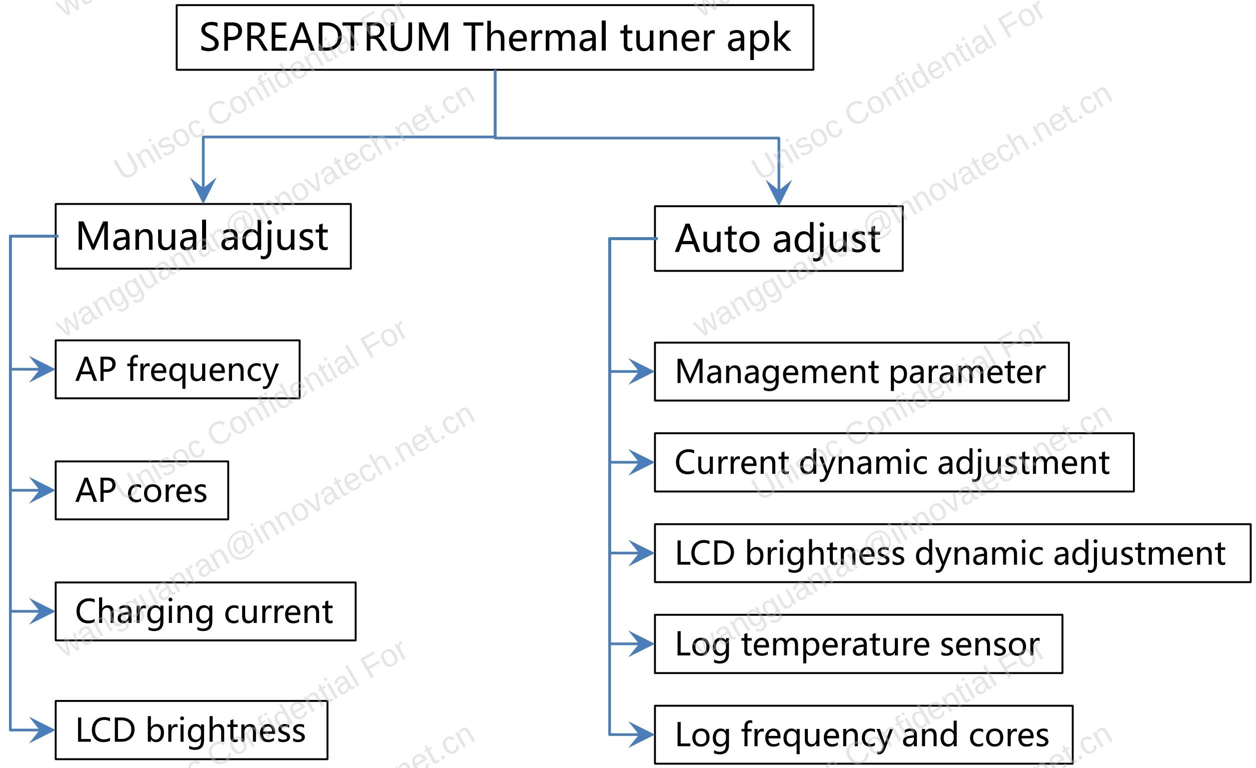
Thermal test notes

- Thermal debug procedure



Thermal test notes

- Thermal debug tool

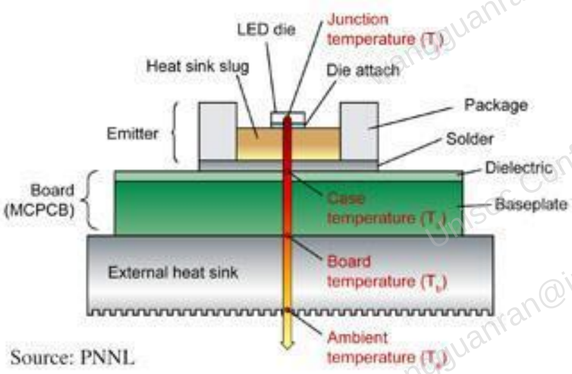


TIM application notes

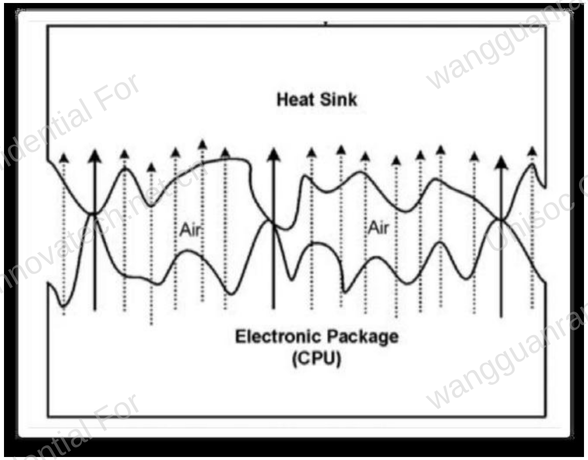
- TIM introduction
- TIM in Consumer Electronics
- Properties of TIM
- Thermal pad introduction
- Graphite sheet introduction
- Graphite sheet application examples
- Graphite sheet application notes
- Nano Silica Baloon Insulator

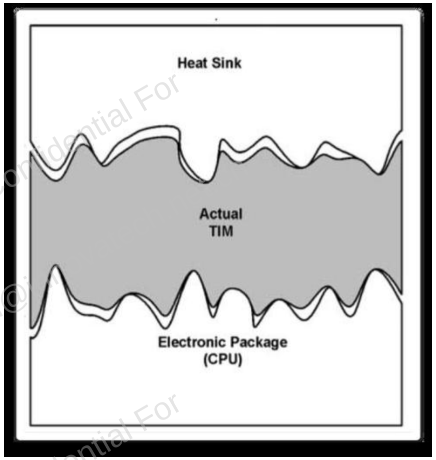
TIM application notes

- TIM introduction



Source: PNNL





➢Overheating is the most critical issue in the computer industry. It limits further miniaturization, power, performance and reliability.

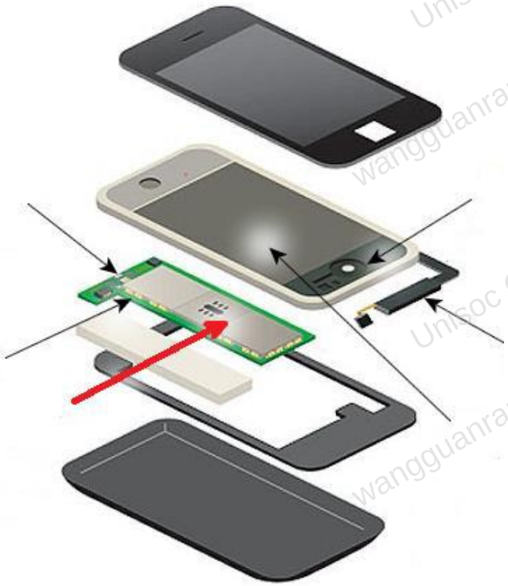
➢Various interfaces exist between high power, heat generating components and heat sinks.

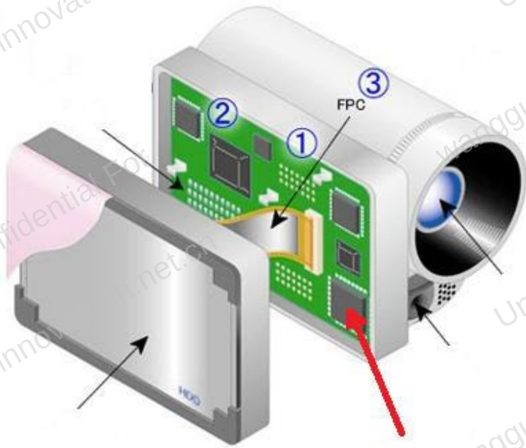
➢Sometimes, there will be only a small contact area between the two surfaces at this interface (sometimes as low as 3%), due to the micro-scale surface roughness. The surface irregularity is the primary cause of thermal contact resistance.

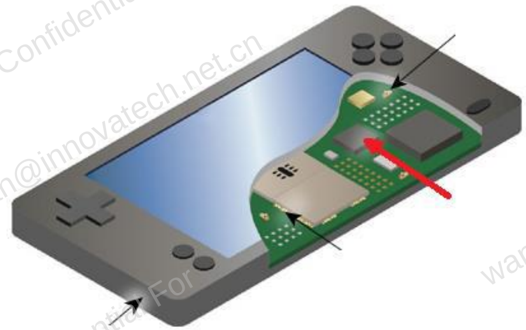
➢Thermal interface materials are required to enhance the contact between the surfaces, and decrease thermal interfacial resistance.

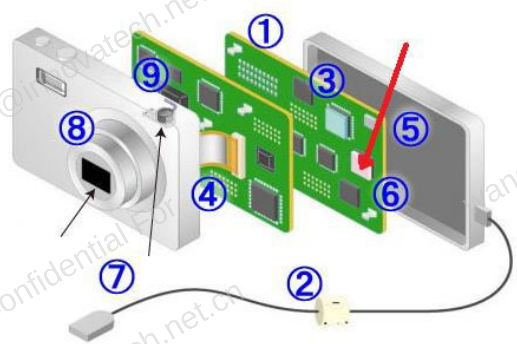
TIM application notes

- TIM in Consumer Electronics









TIM application notes

- Properties of TIM

Property	Tapes	Liquid Adhesives	Greases	Gels and Pastes	Elastomeric Pads	Phase Change Materials	Graphite
Bulk Thermal Conductivity	Moderate	Moderate	Excellent	Good	Excellent	Good	Excellent
Low Interfacial Resistance	Poor	Excellent	Good	Good	Moderate	Excellent	Moderate
Low Bond Line Thickness	Moderate	Good	Poor	Good	Poor	Excellent	Excellent
Application Precision	Excellent	Poor	Poor	Poor	Excellent	Excellent	Good
Ease of Manufacture	Good	Moderate	Poor	Moderate	Good	Good	Good
Longevity	Moderate	Moderate	Poor	Moderate	Moderate	Moderate	Excellent
Rework ability	Poor	Good	Good	Good	Excellent	Excellent	Poor
Stress Relief	Moderate	Moderate	Poor	Excellent	Good	Good	Moderate
Low cost	Good	Good	Excellent	Good	Good	Moderate	Moderate

● — Excellent ● — Good ● — Moderate ● — Poor

TIM application notes

- Thermal pad introduction

Thermal Pads die cut to a precise shape, ready for assembly.

Advantages

- Clean and easy to handle.
- Can be die---cut in the precise shape needed for the application.
- Simplifies assembly.
- High conductivity.
- High dielectric strength.
- Gap Filling.
- Naturally Tacky.
- Soft/Outstanding compression performance.

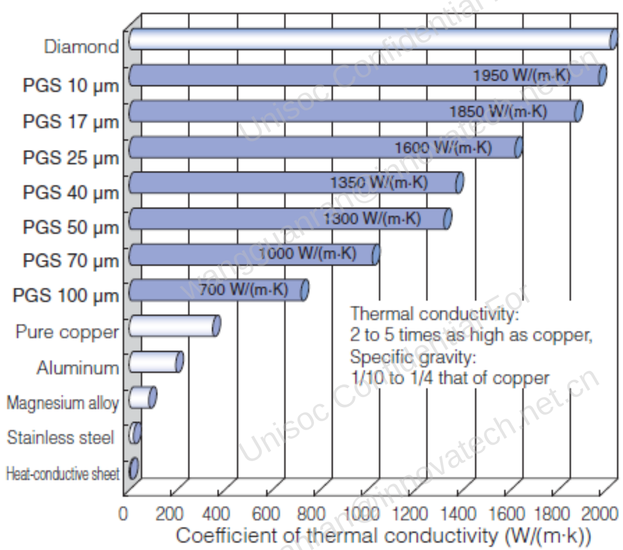
Disadvantages

- To work effectively, thermal pads may require high clamping pressure

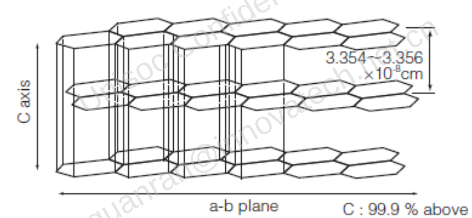
- Graphite sheet introduction

PGS (Pyrolytic Graphite Sheet) is a thermal interface material which is very thin, synthetically made, has high thermal conductivity, and is made from a highly oriented graphite polymer film. It is ideal for providing thermal management/heatsinking in limited spaces or to provide supplemental heat-sinking in addition to conventional means. This material is flexible and can be cut into customizable shapes.

Comparison of thermal conductivity (a-b plane)

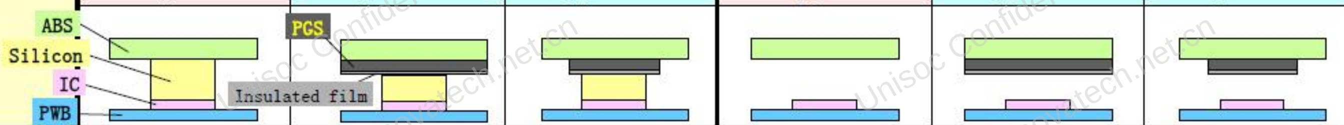
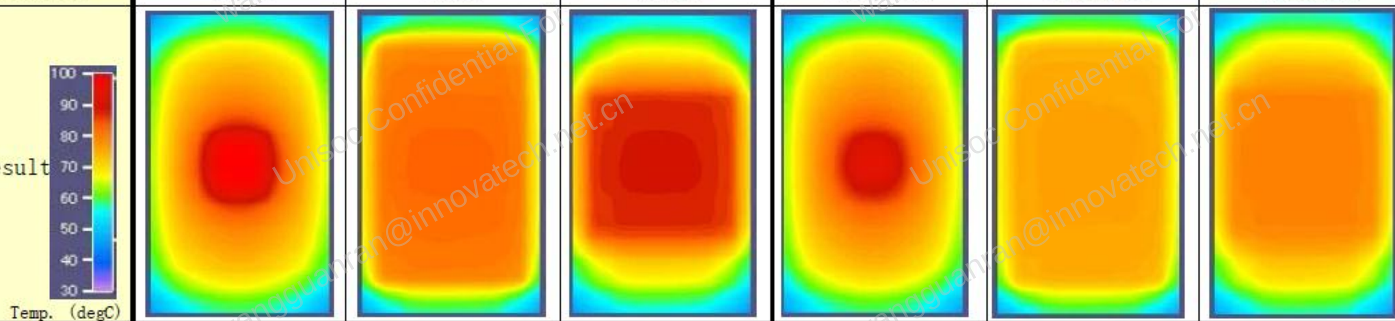


Layered structure of PGS



TIM application notes

- Graphite sheet application examples

		Type A	Type A-1	Type A-2	Type B	Type B-1	Type B-2
Model 							
PGS size (mm)		without	25×40×0.07 (Large)	25×25×0.07 (Small)	without	25×40×0.07 (Large)	25×25×0.07 (Small)
Silicon		with	with	with	without	without	without
Result 							
Temp. (degC)	Surface	99.85	83.84	89.08	93.65	77.17	80.86
	IC	101.9	88.89	93.26	103.2	99.76	100.96
	PWB	96.25	85.31	89.06	97.26	94.19	95.31

TIM application notes

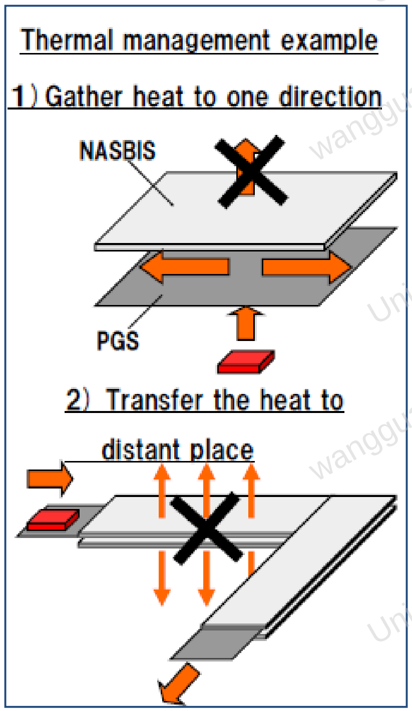
- Graphite sheet application notes

> Graphite sheet is soft, do not rub or touch it with rough materials to avoid scratching it.
> Lines or folds in the Graphite sheet may affect thermal conductivity.
> Graphite sheet has conductivity, should avoid to cover antenna zone.

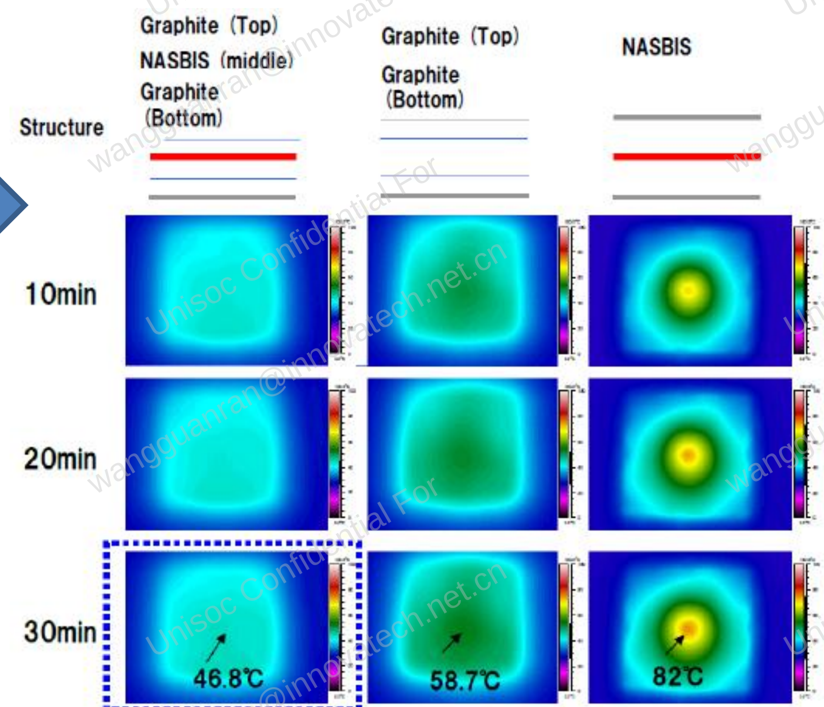
TIM application notes

- Nano Silica Baloon Insulator

  ➢ Combining thermal insulation material with PGS enables thermal management in cellphone.



NASBIS+PGS



THANK YOU!